CLINICAL TRIAL FAILURES · DISCONTINUATION RATE · NEUROLOGY PHASE 2,3 · STARTS 2015–2024

# Amyloid-β directed: 4 molecules stopped early, all monoclonal antibodies

**13 stopped trials, but 4 molecules — all monoclonal antibodies: Gantenerumab, Aducanumab, Crenezumab and Solanezumab.** 3 of the stops were the trial's own verdict and 9 followed a decision taken elsewhere; 1 cannot be established. Against 9.6% for all neurology trials with a resolved drug target and 6.4% across all neurology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

**65.0%**
13 of 20 closed trials (95% CI 43.3%–81.9%)

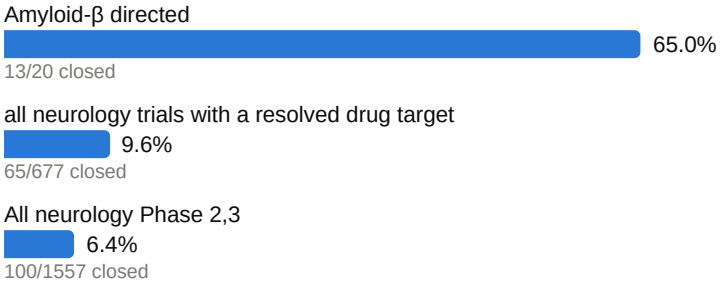
**13 / 0 / 0**
efficacy / safety / both (adds to 13)

**76.9%**
of 26 trials have closed · 6 still open or unresolved

**27.5%**
stopped for a biological reason within 3 years of starting (95% CI 14.2%–49.2%; 15 still at risk; 3.4% area-wide)

**0 of 1** **Completed and missed the primary endpoint.** Of 5 completed trials in this segment, 1 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 0 missed and 1 met. Counted separately and never in the rate above; a floor, not a rate. 1 of the 5 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

## How this compares




**What is in this class:** 7 distinct experimental drugs across 26 trials (antibody 25, unknown 1, small molecule 1). Most tested: GANTENERUMAB (7), DONANEMAB (6), ADUCANUMAB (5), CRENEZUMAB (3), LECANEMAB (3). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

**How much does this rest on one decision?** The 13 stops came from 5 sponsor-asset programmes across 4 sponsors; the largest (Roche) contributed 6; removing that programme's trials from both sides leaves 50.0%. Screened alongside 27 other segments in this area, the rate stays unusual after a 10% false-discovery correction (q=2.5e-07, Benjamini-Yekutieli). A further 2 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 75.0%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

## By start cohort

| Trials started | Stops | Closed | Rate | 95% CI |
|---|---|---|---|---|
| 2015–2017 | 5 | 6 | 83.3% | 43.6%–97.0% |
| 2018–2020 | 6 | 10 | 60.0% | 31.3%–83.2% |

## Where the stops sit

Efficacy / futility: 13.
Sponsors with most stops: Roche (8), Biogen (3), Washington University School of Medicine (1), Eli Lilly (1).

## The stopped trials

| Trial | Ph | Sponsor | Experimental drugs | Stopped | Type | Registry stop reason |
|---|---|---|---|---|---|---|
| NCT06424236 | 3 | Washington University School of Medicine | GANTENERUMAB | 2023-10 | Efficacy | DIAN-TU announced the discontinuation of the DIAN-TU-001 OLE based on the status of the drug program and findings from an interim efficacy analysis and. |

| Trial | Ph | Sponsor | Experimental drugs | Stopped | Type | Registry stop reason |
|---|---|---|---|---|---|---|
| NCT05256134 | 3 | Roche | GANTENERUMAB | 2023-03 | Efficacy | Decision to terminate development of gantenerumab for treatment of people at risk for or at earliest stages of Alzheimer disease (AD) following results of a pre-planned analysis of safety \& efficacy of gantenerumab in Graduate I\&II (WN29922/WN39658). |
| NCT04374253 | 3 | Roche | GANTENERUMAB | 2023-03 | Efficacy | Decision to terminate development of Gantenerumab for treatment of prodromal/mild/early-stage Alzheimer's disease following results of a pre-planned analysis of the safety and efficacy of Gant in Graduate I\&II (WN29922/WN39658). |
| NCT04592341 | 2 | Roche | GANTENERUMAB | 2023-01 | Efficacy | Decision to terminate development of Gantenerumab for treatment of prodromal/mild/early-stage Alzheimer's disease following results of a pre-planned analysis of the safety and efficacy of Gant in Graduate I\&II (WN29922/WN39658). |
| NCT04339413 | 3 | Roche | GANTENERUMAB | 2023-01 | Efficacy | Decision to terminate development of Gantenerumab for treatment of prodromal/mild/early stage Alzheimer's disease following results of a pre-planned analysis of the safety and efficacy of Gant in Graduate I\&II (WN29922/WN39658). |
| NCT03444870 | 3 | Roche | GANTENERUMAB | 2022-12 | Efficacy | Decision to terminate development of Gantenerumab for treatment of prodromal/mild/early-stage Alzheimer's disease following results of a pre-planned analysis of the safety and efficacy of Gant in Graduate I\&II (WN29922/WN39658). |

The 6 most recent of 13 stopped trials. Every molecule behind all 13 is named above; the complete trial list, the trials still running and the terminations with no readable cause ship with the dataset and the diligence report.

**Method**

Denominator: ClinicalTrials.gov interventional Phase 2,3 neurology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 45% of industry neurology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

**Limits**

Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 76.9% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

**The whole cohort, trial by trial.** The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. Free during the beta: enter your work email at **clinicaltrialfailures.com/asset-check** and it opens straight away.

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.